
ME 308: Manufacturing Engineering

Course Code	ME 308
Course Name	Manufacturing Engineering
L-T-P-C	3-0-0-3
Prerequisites	IC 141 Product Realization Technology
Intended for	B.Tech.
Distribution	Core
One-time approval	10th Senate (Jun 2016)

Course Contents

- **Sheet Metal Working**

Types of presses, Operations (shearing, bending, spinning, embossing, blanking, coining, punching and deep drawing), Design of structures using sheet metal working. [7 Lectures]

- **Introduction to Jigs and Fixture Design**

Principles of location and clamping [3 Lectures]

- **Non-conventional Machining Processes**

Electric discharge machining (EDM), Electrochemical machining, LASER and Abrasive flow machining. [8 Lectures]

- **Introduction to CIM**

Trends in Modern Manufacturing, Techniques to enhance flexibility, productivity, product quality and interoperability, Product life cycle, Concepts of product development, Building blocks of CIM. [8 Lectures]

- **Rapid prototyping**

Need for Rapid Prototyping, Basic Principles and advantages of RP, Classifications of different RP techniques with examples, Introduction to three representative RP techniques: Fused deposition modeling, Laminated object manufacturing and Stereo-lithography [8 Lectures]

- **Micro-manufacturing**

An overview of micro mechanical systems and their applications, MEMS Microfabrication methods, Silicon Micromachining methods, Laser Micromachining methods, Mechanical Micromachining techniques, CAD/CAM Tools for Micro-manufacturing processes. [8 Lectures]

Text Books

1. Serope Kalpakjian, and Steve R. Schmid, **Manufacturing Engineering and Technology**, 4th Edition, Pearson, 2016